

Course Outline			
School / Department	School of Systems and Technology – SST / Department of Computer Science		
Course Code	CC-141		
Course Title	Discrete Structures		
Theory / Lab / FYP	Theory		
Degree Program	BS-CS		
Credit Hours	3+0	Contact Hours	3
Pre-requisite	None		
Teaching Methodology	Classroom Lectures, Assignments, Class Activities, Presentations, Quizzes, Mid Term Exam, Final Term Exam.		
Methods of Assessment (Can be changed as per course requirement)	Assignment(s)	10%	
	Quizzes	15%	
	Mid-Term Examination	25%	
	Class Activities	10%	
	Final Examination	40%	
	Total	100%	
Course Moderator / Coordinator	Dr. Iqra Javed		
Contact	Iqra.javed@umt.edu.pk		
Counseling Hours	Monday - Friday (10:00 am – 11:00 am)		
Semester Offered	1 st		
Course Synopsis	Discrete Structures course provides students with a foundational understanding of mathematical concepts and structures that are crucial in computer science and related fields. Topics typically covered include set theory, logic, proof techniques, combinatorics, graph theory, trees, and relations. Through this course, students develop analytical and problem-solving skills, enabling them to tackle complex computational problems, design algorithms, and reason rigorously about discrete systems, forming a solid basis for further study in computer science and related disciplines.		
Course Objectives	- To teach students how to use logic and formal proof techniques to solve problems and prove theorems in discrete mathematics. - To develop problem-solving skills in discrete mathematics, which are essential for computer science students. - To teach students how to model real-world problems using discrete mathematics concepts and apply problem-solving strategies to solve them.		

Course Learning Outcomes (CLOs)	Domain & BT* Level
After the successful completion of course, the students will be able to:	
CLO-1: Understand the key concepts of Discrete Structures such as Logic, Arguments, Sets, Permutations, Relations, Graphs, and Trees etc.	Cognitive, [C2 – Understand]
CLO-2: Apply the concepts of Discrete Structures to solve real world problems and other computing problems.	Cognitive, [C3 – Apply]
CLO-3: Analyze various discrete structures and their relevance within the context of computer science such as Graphs and Trees.	Cognitive, [C4 - Analyze]
* BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain	

Mapping of CLOs to Program Learning Outcomes (PLOs)			
CLOs / PLOs	CLO 1	CLO 2	CLO 3
PL01: Academic Education			
PL02: Knowledge for Solving Computing Problems	✓	✓	✓
PL03: Problem Analysis			
PL04: Design/ Development of Solutions			
PL05: Modern Tool Usage			
PL06: Individual and Teamwork			
PL07: Communication			
PL08: Computing Professionalism and Society			
PL09: Ethics			
PL010: Life-long Learning			

Tentative Week-wise list of topics				
Week	Topics	Reference	Sessional Assessment	CLOs
Week 1	Introduction - Teaching & assessment methodology - Distribution of Course Outlines - Setting up of Norms - Introduction to Discrete Structures Sets and its Operations - Set Theory - Set, Subsets - Cardinality - Cartesian Products - Set Operations - Venn Diagrams	Chapter 2	Class Activity	CLO 1
Week 2	Sequences - Arithmetic and Geometric Sequences - Recurrence Relations	Chapter 2	Class Activity	CLO 2
Week 3	Summations - Summations and Products of Sequences - Double Summations	Chapter 2	Class Activity	CLO 2
Week 4	Foundation of Logic - Logic and Propositional Logic - Compound Proposition and Logical Operators - Logic and bit Operations	Chapter 1	Quiz 1	CLLO 1
Week 5	Propositional Equivalence - Logical Equivalence and its laws - Application of Propositional Logic	Chapter 1	Class Activity	CLO 1, 2
Week 6	Predicate Calculus - Predicates - Quantifiers and its types - Negating Quantified Expressions - Translating from English into Logical Expressions	Chapter 1	Class Activity	CLO 1, 2
Week 7	Rules of Inference - Rules of Inference for Propositional Logic and Quantifiers - Building Arguments - Problem Solving using Rules of Inference	Chapter 1	Quiz 2	CLO 2
Week 8	Proofs - Introduction with proofs - Formal Proofs - Direct and Indirect Proofs	Chapter 1	Assignment 1	CLO 2
Week 9	MID TERM EXAM			CLO 1,2

Tentative Week-wise list of topics				
Week	Topics	Reference	Sessional Assessment	CLOs
Week 10-11	Number Theory - Divisibility and Modular Arithmetic - Primes and Greatest Common Divisors - Solving Convergences - Cryptography	Chapter 4	Class Activity	CLO 1, 2
Week 12	Mathematical Induction - Simple and Strong Induction - Program Correctness	Chapter 5	Quiz 3	CLO 2
Week 13	Functions - Functions and its types - Graphs of Functions - Inverse Functions - Compositions of Functions	Chapter 2	Class Activity	CLO 1, 2
Week 14	Relations - Relations and their Properties - Functions as Relations - Relations on Sets - Combining Relations	Chapter 9	Quiz 4	CLO 2
Week 15	Counting - Basic rules - Pigeonhole Principle - Permutations and Combinations - Binomial Coefficients	Chapter 6	Assignment 2	CLO 2
Week 16	Graphs - Graph terminologies - Graph types - Representing graphs - Connectivity - Shortest Path Problems	Chapter 10	Quiz 5	CLO 3
Week 17	Revision			
Week 18	FINAL TERM EXAM			CLO 1,2,3
Textbook	Discrete Mathematics and Its Applications by Kenneth H. Rosen (Latest Edition)			
Other References	- Discrete Mathematics with Applications by Susanna S. Epp - Discrete Mathematics by Richard Johnsonbaugh			

Mapping of CLOs to Direct Assessments									
CLOs	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Assignment 1	Assignment 2	Mid Term	Final Term
1	✓					✓		✓	✓
2		✓	✓	✓		✓	✓	✓	✓
3					✓		✓		✓

Course Moderator Name: Dr. Iqra Javed

Course Moderator Sign.: _____